

1 Principles of Monte Carlo

1.1 Introduction

Monte Carlo integration

Consider estimating $\alpha = \int_0^1 f(x) dx$. If $U \sim \text{Unif}(0, 1)$, then $\alpha = \mathbb{E}[f(U)]$. Given independent samples $U_1, \dots, U_n \sim \text{Unif}(0, 1)$, the Monte Carlo estimator is

$$\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n f(U_i).$$

By the strong law of large numbers, $\hat{\alpha}_n \rightarrow \alpha$ a.s. as $n \rightarrow \infty$. By the central limit theorem,

$$\hat{\alpha}_n - \alpha \approx N(0, \sigma_f^2/n),$$

where $\sigma_f^2 = \text{var}(f(U))$. Thus the standard error is $\sigma_f/\sqrt{n}$. Halving the standard error requires increasing the number of samples by a factor of 4.

Dimensionality

For one-dimensional integration, Monte Carlo is usually not competitive with deterministic methods such as the trapezoidal rule, whose error is $O(n^{-2})$. The advantage of Monte Carlo is that its convergence rate is not directly sensitive to the dimension of the problem, while many deterministic integration methods suffer from the curse of dimensionality.

1.2 First Example: European Call Option

Risk-neutral valuation

Let $S(t)$ be the stock price at time t . A European call option with strike K and maturity T has payoff $(S(T) - K)^+ = \max(0, S(T) - K)$. Under the risk-neutral measure $\mathbb{Q}$, the time-0 price is

$$V_0 = \mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (S(T) - K)^+],$$

where r is the continuously compounded risk-free rate.

Stock-price dynamics

Under $\mathbb{Q}$, geometric Brownian motion satisfies $\frac{dS(t)}{S(t)} = r dt + \sigma dW(t)$, where $W(t)$ is a standard Brownian motion. The exact solution at time T is

$$S(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma W(T)\right).$$

Since $W(T) = \sqrt{T}Z$ with $Z \sim N(0, 1)$,

$$S(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\right).$$

Simulation algorithm

For $i = 1, \dots, n$, generate $Z_i \sim N(0, 1)$, then set

$$S_i(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i\right),$$

$$C_i = e^{-rT} (S_i(T) - K)^+.$$

The Monte Carlo estimator is $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$. It is unbiased, $\mathbb{E}[\hat{C}_n] = C$, and strongly consistent, $\hat{C}_n \rightarrow C$ with probability 1.

Confidence interval

The sample standard deviation of the simulated discounted payoffs is

$$s_C = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (C_i - \hat{C}_n)^2}.$$

An asymptotic $1 - \delta$ confidence interval is

$$\hat{C}_n \pm z_{\delta/2} \frac{s_C}{\sqrt{n}}.$$

Since s_C is estimated, one may replace the normal quantile with the corresponding t -quantile with $n - 1$ degrees of freedom.

Black-Scholes benchmark

The Black-Scholes value of a European call is

$$BS(S, \sigma, T, r, K) = S\Phi(d_1) - e^{-rT} K\Phi(d_2),$$

where

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}},$$

$$d_2 = \frac{\log(S/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

1.3 Simulation Layers

Layered construction

Monte Carlo derivative pricing is built in layers:

$$U_1 \rightarrow Z_1 \rightarrow S_1(T) \text{ or } \{S_i(t_j)\}_{j=1}^{m-1} \rightarrow C_i \rightarrow \hat{C}_n, s_C.$$

The lower layers transform simple random variables into asset prices or paths. The upper layers transform paths into discounted payoffs and then into statistical summaries.

1.4 Second Example: Asian Option

Path dependence

An Asian option is path dependent. A discrete-average Asian call has payoff $(\bar{S} - K)^+$, where

$$\bar{S} = \frac{1}{m} \sum_{j=1}^m S(t_j),$$

and $0 = t_0 < t_1 < \dots < t_m = T$. Its time-0 value is

$$\mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (\bar{S} - K)^+].$$

To estimate this value, simulate the path values $S(t_1), \dots, S(t_m)$, compute the average $\bar{S}$, discount the payoff, and average over independent replications.

1.5 Efficiency of Simulation Estimators

Central limit theorem

Suppose $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$, where C_i are i.i.d., $\mathbb{E}_0^{\mathbb{Q}}[C_i] = C$, and $\text{var}(C_i) = \sigma_C^2 < \infty$. Then

$$\frac{\hat{C}_n - C}{\sigma_C/\sqrt{n}} \approx N(0, 1),$$

or equivalently

$$\sqrt{n}(\hat{C}_n - C) \approx N(0, \sigma_C^2).$$

Hence $\hat{C}_n - C \approx N(0, \sigma_C^2/n)$. For two unbiased estimators of the same quantity, the estimator with lower variance is preferred, other things being equal.

1.6 Bias

General principle

Variance reduction and faster computation are useful only if the estimator converges to the correct value. Some estimators are biased for every finite n , but become asymptotically unbiased as computational effort increases.

Ratio estimator

For i.i.d. samples (X_i, Y_i) , the estimator $\bar{X}/\bar{Y}$ for $\mathbb{E}[X]/\mathbb{E}[Y]$ is generally biased because

$$\mathbb{E}\left[\frac{\bar{X}}{\bar{Y}}\right] \neq \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}.$$

However,

$$\frac{\bar{Y}}{\bar{Y}} \rightarrow \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$$

with probability 1, and the normalized error is asymptotically normal. The bias becomes negligible as $n \rightarrow \infty$.

Model discretization error

If the exact geometric Brownian motion transition is replaced by the Euler recursion

$$S((j+1)h) = S(j)h + rS(j)h + \sigma S(j)h\sqrt{h}Z_{j+1},$$

then the simulated path distribution is not exactly the same as that implied by the continuous-time model. Option values estimated from this approximate dynamics are biased.

This model discretization bias can be reduced or eliminated by: (i) using the exact transition when available; (ii) decreasing the time step h , at the cost of more transitions per path.

Payoff discretization error

For a continuously averaged Asian option,

$$\bar{S} = \frac{1}{T} \int_0^T S(t) dt.$$

Even exact simulation at a finite set of dates cannot compute the continuous average exactly. The payoff must be approximated, which creates payoff discretization error. This bias can be made small by using a smaller simulation time step, but computational cost increases.

Model discretization error comes from approximating the underlying dynamics. Payoff discretization error comes from approximating the payoff functional.

1.7 Mean Square Error

Bias-variance decomposition

For an estimator $\hat{\alpha}$ of α , the mean square error is

$$\text{MSE}(\hat{\alpha}) = \mathbb{E}[(\hat{\alpha} - \alpha)^2].$$

It decomposes as

$$\text{MSE}(\hat{\alpha}) = (\mathbb{E}[\hat{\alpha}] - \alpha)^2 + \mathbb{E}[(\hat{\alpha} - \mathbb{E}[\hat{\alpha}])^2],$$

or

$$\text{MSE}(\hat{\alpha}) = \text{Bias}^2(\hat{\alpha}) + \text{Variance}(\hat{\alpha}).$$

Thus MSE balances bias and variance.

2 Mean-variance and Optimization

2.1 Portfolio Constraints

Review: Markowitz Mean-Variance Problem

Recall the standard Markowitz Mean-Variance Problem:

$$\begin{aligned} \min_{w_0, w} \quad & w^T C w \\ \text{s.t.} \quad & w_0 r_f + w^T \mu = z \\ & w_0 + w^T \mathbf{1}_n = 1 \end{aligned}$$

where w_0 is the weight in the risk-free asset, w is the vector of weights in risky assets, C is the covariance matrix, and μ is the expected return vector.

Equivalent formulation:

$$\begin{aligned} \min_w \quad & \frac{1}{2} w^T C w \\ \text{s.t.} \quad & w^T (-r_f \mathbf{1}_n) = z - r_f \end{aligned}$$

The above formulation does not take into accounts of constraints in real trading.

Basic Trading Constraints

No short selling: $w_i \geq 0$, for $i = 1, \dots, n$.

No borrowing: $w_0 \geq 0$, which implies $\sum_{i=1}^n w_i \leq 1$.

Margin Requirement for Short Sale

Suppose that for stock i , the short sale of account of this stock should have $(1 + m_i)$ of the value of the short sale. Suppose the margin account earns a deterministic net return rate r_c . Denote by v_i the amount of margin prepared for the short sale of stock i , and denote $v := (v_1, \dots, v_n)$.

Portfolio Net Return:

$$\begin{aligned} R_{(w,v)} &= \bar{R} + v^T r_c \mathbf{1}_n + (1 - w^T \mathbf{1}_n - v^T \mathbf{1}_n) r_f \\ &\Rightarrow R_{(w,v)} = r_f + w^T (\bar{R} - r_f \mathbf{1}_n) + v^T (r_c - r_f) \mathbf{1}_n \end{aligned}$$

Moments: Variance is given by $\text{Var}(R_{(w,v)}) = w^T C w$. Expectation is given by

$$\mathbb{E}[R_{(w,v)}] = r_f + w^T (\bar{R} - r_f \mathbf{1}_n) + v^T (r_c - r_f) \mathbf{1}_n$$

Constraints: Constraints to model margin requirements:

$$v_i \geq -(1 + m_i) w_i, \quad v_i \geq 0, \quad i = 1, \dots, n.$$

Cardinality Constraints

Due to various forms of market friction, most investors can invest in only a limited number of assets, leading to cardinality constraints. Suppose one can invest in at most s stocks. The cardinality constraint can be formulated as the following linear constraints:

$$-M\delta_i \leq w_i \leq M\delta_i, \quad \delta_i \in \{0, 1\}, \quad i = 1, \dots, n$$
$$\sum_{i=1}^n \delta_i \leq s$$

where M is a given, sufficiently large number, and δ_i denotes whether one invests in stock i .

Buy-In Threshold Constraints

Small positions have very limited impact on the total performance of the portfolio, but trigger non-negligible establishing and maintaining cost. To avoid small positions, one can impose a buy-in threshold constraint to prevent from holding an active position less than a prescribed proportion $w_{\min} > 0$ of the total capital. The buy-in threshold constraint can be modeled as the following linear constraints:

$$w_i \leq M\delta_i, \quad w_{\min} \delta_i \leq w_i, \quad \delta_i \in \{0, 1\}, \quad i = 1, \dots, n$$

where M is a given, sufficiently large number.

Diversification Constraints

Suppose that the n stocks are divided into K sectors, and S_k is the collection of the identities of stocks in section k , $k = 1, \dots, K$. Suppose that the investor wants to invest in at least $K_{\min}$ sectors, and for each sector she wants to invest, the minimum predefined level of investment is $s_{\min}$. The above diversification constraint can be modeled as the following linear constraints:

$$\sum_{k=1}^K \delta_k \geq K_{\min}$$
$$s_{\min} \delta_k \leq \sum_{i \in S_k} w_i \leq M \delta_k, \quad k = 1, \dots, K$$

where M is a given, sufficiently large number.

Proportional Transaction Costs and Bid-Ask Spread

Suppose that there is a proportional transaction cost that amounts to $c_{x,b}$ dollars for each dollar invested in a stock, and a similar cost $c_{x,s}$ when selling the stock at the end of the period. For simplicity, suppose short sales are not allowed. Let w_t represent the proportion of the initial capital, pre-transaction cost, invested in stock i . Proportional transaction costs can be taken into account by modifying the net return rate of a stock R_i , which is computed from the stock price directly, as follows:

$$R'_i := \frac{1 - c_{x,b}}{1 + c_{x,b}} (1 + R_i) - 1$$

To take into account the bid-ask spread, one only needs to use the ask price at the beginning of the period and the bid price at the end of the period to compute the net return rate R'_i .

2.2 Relative Entropy Model

Suppose that the agent is uncertain about the model, which is represented by the probability density function of the excess return of the risky assets. The agent has a reference model f_0 but considers alternative models f . Consider the log likelihood ratio of an alternative model f and the reference model f_0 based on observed asset returns X_t :

$$R(X_1, \dots, X_T) = \sum_{t=1}^T \log(f(X_t)/f_0(X_t))$$

By the law of large numbers, under model f :

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T} R(X_1, \dots, X_T) &= \mathbb{E}[\log(f(X)/f_0(X))] \\ &= \mathbb{E}_0[\log(f(X)/f_0(X))] \end{aligned}$$

The term $\mathbb{E}_0[\log(f(X)/f_0(X)) \log(f(X)/f_0(X))]$ is called the relative entropy of the model f with respect to the reference model f_0 .

The agent faces the following problem:

$$\min_{f \in \mathcal{P}} \max_{X'} \left[\lambda w^T \mathbb{E}'[X] - \frac{1}{2} w^T \text{Var}'(X|w) \right]$$

where X stands for the excess return vector of the risky assets and Var' stands for the variance under f , and

$$\mathcal{P} := \{f \mid \mathbb{E}_0[\log(f(X)/f_0(X)) \log(f(X)/f_0(X))] \leq \eta\}$$

for some positive parameter η .

Assume that the excess return vector follows joint normal distribution. Assume that the covariance matrix C is known, and alternative model f differs from the reference model f_0 in the mean. Denote the mean under f and f_0 as μ and μ_0 , respectively. Then, the relative entropy of f is:

$$\mathbb{E}_0[\log(f(X)/f_0(X)) \log(f(X)/f_0(X))] = \frac{1}{2} (\mu - \mu_0)^T C^{-1} (\mu - \mu_0)$$

The agent's problem becomes:

$$\min_{\mu} (\mu - \mu_0)^T C^{-1} (\mu - \mu_0) \leq 2\eta \quad \max_{X'} \left[\lambda w^T \mu - \frac{1}{2} w^T C w \right]$$

Optimal solution:

If $\mu_0^T C^{-1} \mu_0 \leq 2\eta$, then $w^* = 0$.

If $\mu_0^T C^{-1} \mu_0 > 2\eta$, then $w^* = \lambda \left(1 - \sqrt{\frac{2\eta}{\mu_0^T C^{-1} \mu_0}} \right) C^{-1} \mu_0$

2.3 A Smooth-Ambiguity Model

The worst-case mean-variance portfolio selection is usually too conservative. We consider the smooth-ambiguity model, in which the investor has a belief of the model/parameters for the stock distributions, considers the average model/parameter setup instead of the worst-case scenario, and is averse to ambiguity about the model/parameters.

Formally, denote by $P = (M, C)$ a model specification. The investor has a belief of P , represented by distribution π . Let $\bar{P} = (\bar{M}, \bar{C}) = \mathbb{E}_{\pi}[P]$ be the average model specification. The portfolio selection can be formulated as:

$$\max_w \mathbb{E}_{\pi}[R_w] - \frac{\lambda}{2} \text{Var}_{\pi}(R_w) - \frac{\theta}{2} \text{Var}_{\pi}(\mathbb{E}_{\pi}[R_w])$$

where $\lambda > 0$ is the risk aversion parameter and $\theta > 0$ is the ambiguity aversion parameter.

3 Mean-Risk and EU Max

Denote by $R = (R_1, \dots, R_n)^T$ the net return vector of the stocks and r_f the risk-free net return rate. Given portfolio $w = (w_1, \dots, w_n)^T$, where w_i represents the percentage of initial wealth invested in stock i , the net return rate of the portfolio is:

$$R_w = r_f + w^T (\bar{R} - r_f \mathbf{1}_n)$$

3.1 Mean-Semi-variance Model

One alternative to variance is semi-variance:

$$\mathbb{E}[(\min(R_w - \mathbb{E}[R_w], 0)^2)]$$

This measures the downside risk of the portfolio. If the distribution is symmetrical, semi-variance is proportional to variance; in general, it is not the case. The Mean-semi-variance portfolio selection problem is formulated as:

$$\begin{aligned} \min_w \quad & \mathbb{E}[(\min(R_w - \mathbb{E}[R_w], 0)^2)] \\ \text{s.t.} \quad & \mathbb{E}[R_w] \geq z \end{aligned}$$

3.2 Mean-Absolute Deviation Model

Another alternative to variance is absolute deviation:

$$\begin{aligned} \min_w \quad & \mathbb{E}[|R_w - \mathbb{E}[R_w]|] \\ \text{s.t.} \quad & \mathbb{E}[R_w] \geq z \end{aligned}$$

Mean-absolute deviation portfolio selection is formulated as:

$$\begin{aligned} \min_w \quad & \mathbb{E}[|R_w - \mathbb{E}[R_w]|] \\ \text{s.t.} \quad & \mathbb{E}[R_w] \geq z \end{aligned}$$

3.3 Safety-First Model

Safety-first portfolio selection is formulated as:

$$\begin{aligned} \min_w \quad & \mathbb{P}(R_w \leq D) \\ \text{s.t.} \quad & \mathbb{E}[R_w] \geq z \end{aligned}$$

Here, D is a given threshold below which the portfolio return is considered to be a disaster. Both the mean-semi-variance and the mean-absolute deviation portfolio selection problems are convex programs, but the safety-first portfolio selection problem is not.

3.4 Tail Risk Measures

In the safety-first criterion, investors are concerned about the left tail of the return distribution.

Given initial wealth x_0 and a portfolio w , the wealth at the end of the period is $X_T = x_0(1 + R_w)$, and thus the P&L of the portfolio is:

$$Z := X_T - x_0 = x_0 R_w$$

Definition of VaR

Let Z be a random P&L, with negative values referring to losses and positive values to gains. Given a level $\alpha \in (0, 1)$, informally the α -VaR, denoted as $VaR_{\alpha}(Z)$, is defined by:

$$\mathbb{P}(-Z \leq VaR_{\alpha}(Z)) = \alpha$$

Formally, $VaR_{\alpha}(Z)$ is defined as follows:

$$\begin{aligned} VaR_{\alpha}(Z) &= \inf\{x \in \mathbb{R} : \mathbb{P}(-Z \leq x) \geq \alpha\} \\ &= -\sup\{y \in \mathbb{R} : \mathbb{P}(Z < y) \leq 1 - \alpha\} \end{aligned}$$

Definition of CVaR

Given a level $\alpha \in (0, 1)$, informally the α -CVaR, denoted as $CVaR_{\alpha}(Z)$, is defined by:

$$CVaR_{\alpha}(Z) = \mathbb{E}[-Z \mid -Z \leq VaR_{\alpha}(Z)]$$

Formally, $CVaR_{\alpha}(Z)$ is defined by:

$$CVaR_{\alpha}(Z) = \frac{1}{1-\alpha} \int_{\alpha}^1 VaR_p(Z) dp$$

CVaR can be derived by solving an optimization problem:

$$CVaR_{\alpha}(Z) = \min_b \{b + \frac{1}{1-\alpha} \mathbb{E}[\max(-Z - b, 0)]\}$$

3.5 Coherent Risk Measures

A risk measure ρ of risk Z assigns a numerical value to a random P&L Z . A risk measure ρ is a coherent risk measure, if it satisfies the following properties:

Monotonicity: for any Z_1 and Z_2 with $Z_1 \leq Z_2$, $\rho(Z_1) \geq \rho(Z_2)$.

Translation invariance: for any constant m and any Z , $\rho(Z + m) = \rho(Z) - m$.

Positive homogeneity: for any Z and any positive constant λ , $\rho(\lambda Z) = \lambda \rho(Z)$.

Subadditivity: for any P&Ls Z_1 and Z_2 , $\rho(Z_1 + Z_2) \leq \rho(Z_1) + \rho(Z_2)$.

CVaR is a coherent risk measure. VaR satisfies monotonicity, translation invariance, and positive homogeneity, but not subadditivity; thus, it is not a coherent risk measure.

3.6 Data-Driven Mean-Risk Portfolio Selection</

Derivation of Asset Pricing

Consider asset $\mathbb{R}_i$ with return R_i , and denote its expectation and standard deviation as $\mathbb{E}[R_i]$ and σ_i , respectively. Consider the mean-variance problem with $z = \mathbb{E}[R_i]$. The optimal portfolio is

$$\begin{bmatrix} w_0 \\ (1-w_0)\bar{w}_M \end{bmatrix}$$

with $1-w_0 = \frac{\mathbb{E}[R_i]-r_f}{\mathbb{E}[R_M]-r_f}$.

For any $\epsilon \in \mathbb{R}$, consider the portfolio which is mixture of $(1-\epsilon)$ of the optimal portfolio and ϵ of asset i . This new portfolio is feasible with expected return z and its standard deviation is

$$\begin{aligned} & Var((1-\epsilon)w_0r_f + (1-\epsilon)(1-w_0)R_M + \epsilon R_i) \\ &= (1-\epsilon)^2(1-w_0)^2 Var(R_M) + 2\epsilon(1-\epsilon)(1-w_0)cov(R_i, R_M) + \epsilon^2 Var(R_i) \\ &=: f(\epsilon) \end{aligned}$$

0 is the minimum point of $f(\cdot)$ because $(w_0; (1-w_0)\bar{w}_M)$ is in the minimum-variance set.

Capital Asset Pricing Model (CAPM)

First-order condition:

$$0 = f'(\epsilon) = -2(1-w_0)^2 Var(R_M) + 2(1-w_0)cov(R_i, R_M)$$

leads to:

$$1-w_0 = \frac{cov(R_i, R_M)}{Var(R_M)}$$

Capital asset pricing model (CAPM):

$$\mathbb{E}[R_i] - r_f = \beta_i (\mathbb{E}[R_M] - r_f)$$

where

$$\begin{aligned} \beta_i &= \frac{cov(R_i, R_M)}{Var(R_M)} \\ \mathbb{E}[R] - r_f &= \beta (\mathbb{E}[R_M] - r_f) \end{aligned}$$

draws security market line on $(\beta, \mathbb{E}[R])$ plane.

β_i is referred to as the beta of the asset. $\mathbb{E}[R_i] - r_f$ is termed as expected excess rate of return of asset i ; $\mathbb{E}[R_M] - r_f$ is the expected excess rate of return of market portfolio.

CAPM says expected excess return of an asset is proportional to expected excess return of market portfolio. The proportion is its beta (β).

$\beta > 0$, i.e., positively correlated; positive expected excess return.

$\beta > 0$, i.e., uncorrelated; zero expected excess return.

$\beta < 0$, i.e., negatively correlated; negative expected excess return.

It is β rather than α that contributes to expected excess return.

Economic Interpretation of CAPM

Consider the portfolio consisting of market portfolio, ϵ of asset i , and $-\epsilon$ of risk-free asset, the last two components being a zero cost trading strategy with P&L $\epsilon(R_i - r_f)$. The variance of the portfolio is:

$$Var(R_M) + 2\epsilon cov(R_i, R_M) + \epsilon^2 Var(R_i)$$

When $\epsilon < 1$, second term dominates third term. If $cov(R_i, R_M) = 0$, asset i does not affect the variance, just like risk-free asset; thus it has zero risk premium. If $cov(R_i, R_M) > 0$, asset i increases the variance; thus it has positive risk premium. If $cov(R_i, R_M) < 0$, asset i reduces the variance; thus it has negative risk premium.

Key insight: risk is the variance of the portfolio return, not the variance of any single asset's return.

Decompose the Risk

For any asset i with (random) return R_i , define ϵ_i via:

$$R_i - r_f = \beta_i(R_M - r_f) + \epsilon_i$$

By CAPM, $\mathbb{E}[\epsilon_i] = 0$. By the definition of β_i :

$$\begin{aligned} cov(R_M, \epsilon_i) &= cov(R_M, R_i - r_f - \beta_i(R_M - r_f)) \\ &= cov(R_M, R_i - r_f) - cov(R_M, \beta_i R_M - \beta_i r_f) \\ &= cov(R_M, R_i) - \beta_i cov(R_M, R_M) = 0 \end{aligned}$$

The risky return R_i can be decomposed into two parts: the part comoves with market return and the part uncorrelated with market return. As a result:

$$\sigma_i^2 = cov(\beta_i(R_M - r_f) + \epsilon_i, \beta_i(R_M - r_f) + \epsilon_i) = \beta_i^2 \sigma_M^2 + Var(\epsilon_i)$$

The first term, $\beta_i^2 \sigma_M^2$ is called **systematic risk**. The second term, $Var(\epsilon_i)$, is called **nonsystematic risk**, **idiosyncratic**, or **specific risk**. Only the systematic risk contributes to expected return according to CAPM. Any asset or portfolio on capital market line (efficient frontier) has only systematic risk. Any asset or portfolio below capital market line has nonsystematic risk.

CAPM as a Pricing Formula

Certainty equivalent pricing formula: The price P of an asset with payoff Q is:

$$P = \frac{1}{1+r_f} \frac{\mathbb{E}[Q] - \frac{cov(Q, R_M)(\mathbb{E}[R_M] - r_f)}{Var(R_M)}}{\mathbb{E}[Q]}$$

Pricing form of CAPM: The price P of an asset with payoff Q (random variables) is:

$$P = \frac{\mathbb{E}[Q]}{1+r_f + \beta(\mathbb{E}[R_M] - r_f)}$$

First formula is used when the correlation of future payoff with market portfolio is known. Second formula is used when β is known a priori.

Example: The Oil Venture

Example 7.6 (The oil venture) Consider the possibility of investing in a share of a certain oil well that will produce a payoff that is random. The expected payoff is \$1,000 and the standard deviation of return is a relatively high 40%. The beta of the asset is $\beta = .6$. The risk-free rate is $r_f = 10\%$, and the expected return of the market portfolio is .17. What is the value of this share of the oil venture, based on CAPM? From the pricing formula, we have:

$$P = \frac{\mathbb{E}[Q]}{1+r_f + \beta(\mathbb{E}[R_M] - r_f)} = \frac{\$1,000}{1+0.1+0.6 \times (0.17-0.1)} = \$876$$

Example: ABC Fund Analysis

Example 7.4 (ABC fund analysis) The ABC mutual fund has a 10-year record of rates of return. We would like to evaluate this fund's performance in terms of mean-variance portfolio theory and the CAPM.

Step 1 Estimate $\mu := \mathbb{E}[R_i]$, r_f , $\mu_M := \mathbb{E}[R_M]$, and β from the data:

$$\begin{aligned} \hat{\mu} &= \frac{1}{n} \sum_{i=1}^n R_i^t, \quad \hat{\mu}_M = \frac{1}{n} \sum_{i=1}^n R_M^t, \quad \hat{r}_f = \frac{1}{n} \sum_{i=1}^n r_f^t \\ cov(R, R_M) &= \frac{1}{n-1} \sum_{i=1}^n (R_i^t - \hat{\mu})(R_M^t - \hat{\mu}_M) \\ Var(R_M) &= \frac{1}{n-1} \sum_{i=1}^n (R_M^t - \hat{\mu}_M)^2 \\ \hat{\beta} &= \frac{cov(R, R_M)}{Var(R_M)} \end{aligned}$$

Step 2 We write the formula:

$$\hat{\mu} - \hat{r}_f = \hat{\beta} (\hat{\mu}_M - \hat{r}_f)$$

If ABC fund is priced according to CAPM, J should be zero statistically. J is known as Jensen's index or alpha (α). For ABC fund, $J > 0$. If you are a CAPM believer, you may think ABC fund is underpriced and thus a good asset. If you are a market efficiency believer, you may think CAPM is invalid.

Step 3 Find the estimate of the Sharpe ratio:

$$S := \frac{\hat{\mu} - \hat{r}_f}{\hat{\sigma}}$$

where

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (R_i^t - \hat{\mu})^2$$

For ABC fund, $S = 0.436$. For S&P 500, $S = 0.467$. If you are a CAPM believer, you may think ABC fund is not efficient.

Testing CAPM Using Linear Regression

Suppose there are n risky assets in the market with return rates in the following period to be R_i , $i = 1, \dots, n$. Suppose the returns of the risk-free asset and of the market portfolio in the following period are r_f and R_M , respectively.

Decomposition:

$$R_i - r_f = \alpha_i + \beta_i(R_M - r_f) + \epsilon_i, \quad i = 1, \dots, n$$

where ϵ_i are independent of each other and of R_M .

Vector form:

$$R - r_f \mathbf{1}_n = \alpha + \beta(R_M - r_f) + \epsilon$$

where $R := (R_1, \dots, R_n)^T$, $\alpha := (\alpha_1, \dots, \alpha_n)^T$, $\beta := (\beta_1, \dots, \beta_n)^T$, and $\epsilon := (\epsilon_1, \dots, \epsilon_n)^T$. ϵ is independent of R_M . CAPM holds if and only if $\alpha = 0$. To test whether CAPM holds, suppose we have historical data of the asset returns: R_i , R_M , r_f , $t = 1, \dots, T$. Suppose the asset returns in different periods are i.i.d. Run vector linear regression to estimate α and β . Reject CAPM if the estimate of α is significantly different from zero.

4.2 Arbitrage Pricing Theory

A Multi-Factor Model

Fix one period and consider n zero-cost strategies in this period with payoffs $F_1, \dots, F_n$, respectively. Assume the following factor model:

$$F_i = \alpha_i + \sum_{k=1}^m \beta_{i,k} f_k + \epsilon_i, \quad i = 1, \dots, n$$

where f_k , $k = 1, \dots, m$ are m common factors that are payoffs of m zero-cost strategies. ϵ_i represent idiosyncratic risk, i.e., $\mathbb{E}[\epsilon_i] = 0$, $cov(\epsilon_i, f_k) = 0$, and $cov(\epsilon_i, \epsilon_j) = 0$ $i = 1, \dots, n$, $j = 1, \dots, n$, $k = 1, \dots, m$, and $i \neq j$. $\beta_{i,k}$ is the loading of asset i on factor k . α_i are constants.

Example: CAPM ($m = 1$). $f = R_M - r_f$, $\beta_i = R_i - r_f$, $\alpha_i = 0$.

Example: Fama-French three-factor model ($m = 3$). $f_1 = R_M - r_f$, f_2 , usually denoted as SMB, is the difference in the returns of two portfolios that are related to the so-called size premium. f_3 , usually denoted as HML, is the difference in the returns of two portfolios that are related to the so-called value premium.

Vector form of the factor model:

$$F = \alpha + \beta \mathbf{f} + \epsilon$$

where $\mathbf{f} := (f_1, \dots, f_m)^T$, $\beta := (\beta_{i,j})$, $\epsilon := (\epsilon_1, \dots, \epsilon_n)^T$, $\alpha := (\alpha_1, \dots, \alpha_n)^T$.

Arbitrage Pricing Theory (APT)

Theorem (Arbitrage Pricing Theory (APT)): Assume there is no arbitrage opportunity in the market. Then, $\alpha = 0$. In consequence,

$$\mathbb{E}[F] = \beta \mathbb{E}[\mathbf{f}]$$

$\mathbb{E}[F]$ is the risk premium of the common factors. The risk premium of asset i is its loading on the common factors multiplied by the risk premia of the common factors.

In CAPM, the only factor is the excess return of the market portfolio, $R_M - r_f$. The risk premium of the common factor is the expected excess return of the market portfolio, $\mathbb{E}[R_M] - r_f$. The loading of the asset i on the common factor is its beta.

Statistical Testing of APT

Suppose we have historical data of the payoffs of the n zero-cost strategies: $\tilde{r}_i$, $t = 1, \dots, T$ and the historical data of the payoffs of the m factors: f_i , $t = 1, \dots, T$. Suppose the payoffs are i.i.d. across periods. Run multiple linear regression to estimate α and β . Reject APT if the estimate of α is significantly different from zero.

5 Multi-Period Investment Models

5.1 Multi-Period Investment

There are N periods of investment. Denote the beginning and end of j -th period as t_{j-1} and t_j , $j = 1, \dots, N$. There is a risk-free asset and m risky stocks in the market. The net return rate of investing in the risk-free asset in the j -th period is a constant $r_{f,j}$. The excess return rate of investing in stock i in period j is $R_{i,j}$.

Assume that $R_j := (R_{1,j}, \dots, R_{m,j})^T$, $j = 1, \dots, N$ are independent.

Consider an agent with initial wealth x_0 at the beginning of first period. The agent's decision is the allocation of his money in each period.

5.2 Wealth Dynamics

Denote as X_j the wealth of the agent at time t_j , $j = 0, 1, \dots, N$. Suppose that except for the initial investment, the agent does not inject new money nor withdraw money from the investment account before the terminal time $T := t_N$.

At the beginning of each period j , the agent decides to invest $\pi_{i,j-1}$ dollars in stock i , $i = 1, \dots, m$, and the remaining amount of money in the risk-free asset. Wealth dynamics:

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

5.3 Expected Utility Maximization

Assume that the agent maximizes the expected utility of her terminal wealth,

$$\max_{\pi_{i,j-1}, j=1, \dots, N} \mathbb{E}[u(X_N)]$$

where u is the agent's utility function and $\pi_{i,j-1} = (\pi_{1,j-1}, \dots, \pi_{m,j-1})^T$, $j = 1, \dots, N$ are the agent's portfolio strategy.

5.4 Dynamic Programming

We apply dynamic programming to solve the problem. Principle of Optimality: Starting from any point on the optimal path, the remaining decisions must constitute an optimal policy with regard to the point.

Consider time t_k when the agent's wealth then is x . Consider the following problem:

$$\max_{\pi_{i,j-1}, j=k+1, \dots, N} \mathbb{E}_k[u(X_N)|X_k = x]$$

Denote the above optimal value function as $v_k(x)$.

Dynamic Programming (DP) Equation:

$$v_k(x) = \max_{\pi_{i,j-1}} \mathbb{E}_k[v_{k+1}(X_{k+1})|X_k = x], \quad k = 0, 1, \dots, N-1$$

$$v_N(x) = u(x)$$

Instead of solving a dynamic portfolio selection problem, we solve a family of single-period portfolio selection problem. $(\pi_0^*, \dots, \pi_{N-1}^*)$ is optimal to the

dynamic portfolio selection problem if and only if π_k^* is optimal to the above single-period portfolio selection problem, $k = 0, \dots, N-1$.

5.5 Power Utility

In the following, we assume power utility function:

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad x > 0$$

for some $0 < \gamma \neq 1$. Applying DP, we have:

$$v_{N-1}(x) = \max_{\pi_{N-1}} \mathbb{E}_{N-1}[u(X_N)|X_{N-1} = x] = b_{N-1}x^{1-\gamma}, \quad x > 0$$

where

$$b_{N-1} = \max_w \frac{\mathbb{E}[(1+r_f+w^T R_{N-1})^{1-\gamma}]}{1-\gamma}$$

Denote the optimizer in the right-hand side of the above as w_{N-1}^* .

By induction, we conclude:

$$v_{k-1}(x) = (\prod_{j=k-1}^{N-1} (b_j(1-\gamma)))^{\frac{1}{1-\gamma}}$$

where

$$b_{k-1} = \max_w \frac{\mathbb{E}[(1+r_f+w^T R_k)^{1-\gamma}]}{1-\gamma}, \quad k = 1, \dots, N$$

Denote the optimizer in the right-hand side of the above as w_{k-1}^* . Then, the optimal portfolio is:

$$\pi_{k-1}^* = w_{k-1}^* X_{k-1}, \quad k = 1, \dots, N$$

Myopic Portfolio

With the power utility function, the investor acts at each stage, based on the optimal policy, as if he were faced with a single-period investment, i.e., as if he were myopic. When the utility function is not a power function or when the stock returns are not independent over time, the above is not necessarily true. For power utility functions, it is implicitly assumed that the wealth must remain positive. Thus, the portfolio π needs to be constrained, i.e., $\pi_k \geq 0$, $i = 1, \dots, n$, and $\sum_{i=1}^n \pi_{i,k} \leq X_k$.

Example: Expected Utility Maximization

Suppose the risk-free net return rate in each period is 0.03, and there is one stock in the market only with i.i.d. excess return rates over time. Suppose that the log return rate of the stock, $\ln(1+r_f+R_k)$ follows normal distribution with mean 0.07 and variance 0.04. Suppose that the investor is not allowed to short-sell the stock or to borrow, i.e., $0 \leq \pi_k \leq X_k$. Suppose that the investor has a power utility function with $\gamma = 10$. To find the optimal portfolio, we need to solve:

$$\max_{\pi_{i,j-1} \in [0,1]} f(w)$$

where

$$f(w) := \frac{\mathbb{E}[(1+r_f+R_k w)^{1-\gamma}]}{1-\gamma}$$

Straightforward calculation yields:

$$f'(w) = \mathbb{E}[(1+r_f+R_k w)^{-\gamma} R_k]$$

By binary search, one can find that $f'(0.1489) = 0$, which implies that the optimal portfolio $w^* = 0.1489$, i.e., it is optimal to invest 14.89

5.6 Long-Run Mean Exponential Growth Rate

Suppose $t_k - t_{k-1} = \Delta$, $k = 1, \dots, N$ in the above, e.g., Δ is one month, so the terminal time $t_N = N\Delta$. Assume that the stock excess returns R_k , $k = 1, 2, \dots$ are i.i.d. For each portfolio strategy $(\pi_1, \pi_2, \dots)$, consider the long-run mean exponential growth rate:

$$\lim_{N \rightarrow +\infty} \mathbb{E} \left[\frac{\log(X_N/X_0)}{N} \right]$$

The objective is to maximize the above long-run mean exponential growth rate.

Kelly's Portfolio

It turns out that the optimal strategy is as follows: at each time $k\Delta$ invest w_k^* proportion of wealth at that time in stock k , $i = 1, \dots, n$, where

$$w_k^* := (w_1^*, \dots, w_n^*)^T = \arg \max_{\pi} \mathbb{E}[\log(1+r_f+w^T R_k)]$$

The above strategy is called Kelly strategy.

Kelly portfolio has many interesting properties: Asymptotically maximizes the rate of asset growth. Asymptotically maximizes the median of wealth. Is a myopic strategy.

The Kelly strategy may lead to a large fraction of wealth in bets or in risky investment. For instance, for a stock with annual mean excess return rate 4% and volatility 20%, the Kelly strategy is to invest 100% of the wealth in the stock.

Fractional Kelly

MacLean et al. (1992) propose fractional Kelly to tradeoff growth and security. κ -fractional Kelly refers to a strategy of investing κ fraction of one's wealth in the Kelly portfolio and holding the remaining as cash (or in the risk-free asset). Fractional Kelly is a heuristic portfolio, rather than the consequence of optimality.

MacLean et al. (1992) lay out two attractive properties of fractional Kelly. It is easily computable. There is an effective tradeoff between growth and security by varying κ .

5.7 Multi-Period Mean-Variance Portfolio

Next, we consider multi-period mean-variance portfolio selection. The market setting is as before. Denote by μ_k and C_k the mean and variance of R_k , respectively.

At time 0, given initial wealth x_0 , the agent considers the following dynamic portfolio selection problem:

$$\begin{aligned} \min_{\pi_0, \dots, \pi_{N-1}} & \frac{1}{2} Var(X_N) \\ \text{s.t.} & \mathbb{E}[X_N] = x_0 \end{aligned}$$

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^n \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

$$X_0 = x_0$$

Lagrange Dual Method

Apply the Lagrange multiplier λ_0 to the constraint for the expected terminal wealth target, and we obtain:

$$\min_{\pi_0, \dots, \pi_{N-1}} \frac{1}{2} \mathbb{E}[(X_N - x_0)^2] - \lambda_0 (\mathbb{E}[X_N] - x_0)$$

subject to

$$\begin{aligned} X_j &= (1 + r_{f,j})X_{j-1} + \sum_{i=1}^n \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N \\ X_0 &= x_0 \end{aligned}$$

Then, we apply dynamic programming to solve the above problem: for each $k = 0, 1, \dots, N-1$ and each x , consider:

$$\begin{aligned} \min_{\pi_k, \dots, \pi_{N-1}} & \frac{1}{2} \mathbb{E}_k[(X_N - x_0)^2 | X_k = x] - \lambda_0 (\mathbb{E}_k[X_N | X_k = x] - x_0) \\ \text{s.t.} & X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^n \pi_{i,j-1} R_{i,j}, \quad j = k$$

Question 2

Consider a wealth manager who is looking into a particular portfolio strategy. She found that the historical estimates of the mean and variance of the portfolio's monthly return are 0.7% and 1.8%, respectively. She also estimated the mean and variance of the market portfolio's monthly return to be 0.6% and 1.2%, respectively, and the correlation between the return rates of the portfolio and the market is 0.9. The monthly risk-free return rate is flat at 0.2% in the past.

(a) According to CAPM, is this strategy a good one? Why?

(b) The director of the wealth manager believes that the Fama-French three-factor model instead of CAPM should be used to assess the portfolio strategy. To this end, the wealth manager estimated the loadings of the portfolio for SMB and HML, which turn out to be -0.3 and 0.7, respectively. She also estimated the expected payoffs for the zero-cost strategies associated with SMB and HML to be 0.4% and 0.3%, respectively. Is this strategy a good one under the Fama-French three-factor model? Why?

Solution 2

(a) We compute β_i first:

$$\beta_i = \frac{\text{cov}(r_i, r_M)}{\text{Var}(r_M)} = \frac{\rho \sigma_i \sigma_M}{\sigma_M^2} = \frac{0.9 \times \sqrt{1.8\%} \times \sqrt{1.2\%}}{1.2\%} = 1.10227$$

As a result, the risk premium of the portfolio strategy implied by CAPM is:

$$\bar{r}_i - r_f = \beta(\bar{r}_M - r_f) = 1.10227 \times (0.6\% - 0.2\%) = 0.44091\%$$

This is lower than the estimated risk premium, namely $0.7\% - 0.2\% = 0.5\%$. So under CAPM, this portfolio strategy is good (it offers a positive alpha).

(b) The loadings of the strategy on the market risk factor, SMB, and HML, are 1.10227, -0.3, and 0.7, respectively. The risk premiums of these three factors are 0.6%-0.2%, 0.4%, and 0.3%, respectively. Thus, the model risk premium of the strategy under the Fama-French three-factor model is:

$$1.10227 \times (0.6\% - 0.2\%) - 0.3 \times 0.4\% + 0.7 \times 0.3\% = 0.531\%$$

This is higher than the estimated risk premium of 0.5%. Thus, this strategy is a bad one when benchmarked to the Fama-French three-factor model.

Question 3

Suppose that David's total wealth is \$40,000, in which his car is worth \$15,000. Suppose his utility function is $u(x) = 2\sqrt{x}$. There is a 5% probability that his car would be stolen and he will lose \$15,000. There is also a 10% probability that his car would be damaged in an accident and he will have to spend \$10,000 to repair it.

(a) What is the Arrow-Pratt absolute risk aversion coefficient of David?

(b) If he wants to buy a car insurance which can fully cover his loss in both cases, i.e., stolen and damaged, what is the maximum amount of money he is willing to pay for the insurance?

Solution 3

(a) The Arrow-Pratt absolute risk aversion coefficient is:

$$r(x) = -\frac{u''(x)}{u'(x)} = -\frac{2 \times (1/2) \times (-1/2)x^{-3/2}}{2 \times (1/2)x^{-1/2}} = \frac{1}{2}x^{-1}$$

(b) The maximum amount of money, A , David is willing to pay is the amount such that David is indifferent between his wealth if he purchases the insurance contract and his wealth if he does not purchase the contract.

If David purchases the insurance contract, his wealth is $40000 - A$ and the utility of the wealth is $2\sqrt{40000 - A}$.

If David does not purchase the insurance contract, the expected utility of his wealth is:

$$2\sqrt{40000 - 15000} \times 0.05 + 2\sqrt{40000 - 10000} \times 0.10 + 2\sqrt{40000} \times 0.85 = 390.4524$$

Setting the expected utility of the wealth in the case in which David purchases the insurance equal to the case in which he does not purchase the insurance, namely:

$$2\sqrt{40000 - A} = 390.4524$$

We derive $A = 1886.731$.

Question 4

Consider the following Limit Order Book (LOB) for a security:

Bid Side (Depth Available at Price): \$1.99 (Depth: 2), \$1.98 (Depth: 3), \$1.97 (Depth: 1), \$1.96 (Depth: 2), \$1.95 (Depth: 4).

Ask Side (Depth Available at Price): \$2.02 (Depth: 1), \$2.03 (Depth: 2), \$2.04 (Depth: 2), \$2.05 (Depth: 3), \$2.06 (Depth: 4).

Current Bid Price: \$1.99. Current Ask Price: \$2.02. Current Mid Price: \$2.005.

(a) Suppose an order arrives to buy 3 units of the security at price \$2.00. Calculate the mid price immediately after the arrival of the order.

(b) Suppose an order arrives to sell 1 unit of the security at price \$2.02. Calculate the mid price immediately after the arrival of the order.

(c) Suppose an order arrives to buy 3 units of the security at the market price. Calculate the mid price immediately after the arrival of the order.

(d) Suppose cancellation of an existing sell orders of size 1 at price \$2.02 arrives. Calculate the mid price immediately after the arrival of cancellation.

Solution 4

(a) The order is a buy limit order at \$2.00. Since $\$2.00 < \text{Ask } (\$2.02)$, it becomes the new best bid. New Bid: \$2.00, Ask remains: \$2.02, Mid Price:

$$\frac{2.02 + 2.00}{2} = \$2.01$$

(b) The order is a sell limit order at \$2.02. Since $\$2.02 > \text{Bid } (\$1.99)$ and \$2.02 is the current Ask, it adds depth to the ask side but does not change the best prices. Bid: \$1.99, Ask: \$2.02, Mid Price remains:

$$\frac{2.02 + 1.99}{2} = \$2.005$$

(c) A market buy order for 3 units will walk up the book. It consumes the 1 unit at \$2.02. It consumes 2 units at \$2.03. The new best Ask becomes \$2.04. Bid remains \$1.99. Mid Price:

$$\frac{2.04 + 1.99}{2} = \$2.015$$

(d) Cancellation of 1 unit at \$2.02. The existing depth at \$2.02 was 1. After cancellation, depth at \$2.02 becomes 0. The next available ask is \$2.03. New Ask: \$2.03. Bid remains \$1.99, Mid Price:

$$\frac{2.03 + 1.99}{2} = \$2.01$$

Question 5

Consider a 3-period mean-variance portfolio selection problem with one risk-free asset and 1 risky stock. Suppose the initial wealth is \$1. Suppose that the risk-free net return in each period is $r_{f,k} = 0.03$. In addition, for $k = 1, \dots, 3$ the expected excess return vector of the stocks is:

$$\mu_k = \mathbb{E}[R_k] = 5\%$$

and variance of the stock excess return rate is:

$$C_k = \text{Var}(R_k) = 0.09$$

Suppose the agent's target is to grow the wealth by 6% in each period. Find the optimal portfolio strategy of the agent.

Solution 5

The number of investment periods is $N = 3$. We first compute D_k :

$$D_k = \mathbb{E}[R_k R_k^T] = C_k + \mu_k \mu_k^T = (5\%)^2 + 0.09 = 0.0925$$

$$\mu_k^T D_k^{-1} \mu_k = (5\%)^2 / 0.0925 = 0.027027028, \quad k = 1, \dots, 3$$

Next, we compute:

$$D_k^{-1} \mu_k = 5\% / 0.0925 = 0.54054054$$

$$s_k = \prod_{j=k}^{N-1} (1 + r_{f,j+1}) = 1.03^{3-k}, \quad k = 0, \dots, 2$$

$$\beta_k = \prod_{j=k}^{N-1} (1 - \mu_{j+1}^T D_{j+1}^{-1} \mu_{j+1}) = 0.972972973^{3-k}, \quad k = 0, \dots, 2$$

The target expected final wealth is $z_0 = 1 \times (1 + 0.06)^3$. (The problem says "grow the wealth by 6% in each period", implying $z_0 = (1.06)^3$).

The optimal strategy formula is $\pi_k^* = -A_k s_k + B_k$.

We calculate the coefficients:

$$A_k := (1 + r_{f,k+1}) D_{k+1}^{-1} \mu_{k+1} = 1.03 \times 0.54054054 = 0.556756757, \quad k = 0, 1, 2$$

$$B_k := A_k s_k^{-1} \frac{z_0 - \beta_0 z_0 s_0}{1 - \beta_0}$$

Substituting the values ($s_0 = 1$, $z_0 = (1.06)^3$, $s_0 = 1.03^3$, $\beta_0 = 0.972973$):

$$B_k = 0.556756757 \times \frac{(1.06)^3 - 0.972972973^3 \times 1.03^3 \times 1}{1 - 0.972972973^3} \times 1.03^{k-3}$$

Thus, the values for B_k are:

$$B_k \approx \begin{cases} 1.91400139, & k = 0, \\ 1.227142143, & k = 1, \\ 1.263956407, & k = 2. \end{cases}$$

The optimal portfolio strategy is:

$$\pi_k^* = -0.556756757 s_k + B_k, \quad k = 0, 1, 2$$